Security Risk Assessment Report

Part 1: Select up to three hardening tools and methods to implement

Three hardening tools the organization can use to address the vulnerabilities found include:

- Implementing multi-factor authentication (MFA)
- Enforcing strong password policies
- Regular firewall configuration and maintenance

MFA requires users to verify their identity using more than one authentication factor before accessing systems. Methods include PIN numbers, authenticator apps, ID cards, and biometric verification.

Password policies can include rules for password length, complexity (uppercase, lowercase, numbers, special characters), and login attempt limits. This will discourage password sharing and reduce brute force risks.

Firewall maintenance ensures that rules are updated regularly to block unauthorized or suspicious traffic. Administrators can deny high-risk IP addresses and adapt rules whenever a new threat is detected.

Part 2: Explain your recommendation(s)

Multi-factor authentication (MFA) adds an extra layer of protection against account compromise. Even if a password is stolen or guessed, attackers cannot access systems without the second factor. This reduces risks of brute force attacks and minimizes the value of shared passwords, since additional authentication is required.

Password policies ensure accounts remain difficult to compromise. By enforcing complexity requirements, password expiration rules, and account lockouts after repeated failed login attempts, the organization can significantly reduce the likelihood of unauthorized access. Policies also discourage employees from reusing or sharing weak passwords.

Firewall maintenance is essential for defending against evolving threats. Firewalls should be checked weekly and updated after every security event. Blocking unnecessary or suspicious traffic reduces exposure to denial-of-service (DoS/DDoS) attacks and other intrusion attempts. Consistent updates ensure the firewall reflects the latest standards and threat intelligence.